Middle East Technical University

Department of Electrical and Electronics Engineering

EE568 – Selected Topics in Electrical Machines

Project 1: Torque in a Variable Reluctance Machine

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GitHub Repository:
https://github.com/odtu/EE568/tree/master/Project1

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1 Introduction

This report presents the analytical modeling and simulation analysis of a variable reluctance machine (VRM) as part of EE568 Project 1. Variable reluctance machines generate torque by minimizing the reluctance of the magnetic circuit; unlike permanent magnet or wound-rotor machines, they require no rotor excitation, making them robust and low-cost.

The main objectives of this project are:

1. Derive analytical expressions for the reluctance, inductance, and torque as a function of rotor position.

2. Model the system in a 2D finite element analysis (FEA) software using linear material properties and compare with the analytical results.

3. Extend the FEA model to non-linear material properties and investigate the effects of saturation and fringing flux.

4. Propose a control strategy to achieve continuous rotor rotation.

The machine geometry for this project is defined on the course GitHub page. Some of the key parameters are summarized in Table 1.

Table 1: Machine Parameters

Parameter	**Symbol**	**Value**
Number of turns	N	300
Coil current (DC)	I	2.5 A
Core depth	d	25 mm
Core width	w_c	15 mm
Air-gap clearance	g	0.5 mm
Coil winding area	A_{coil}	30 mm × 10 mm

2 Q1 - Analytical Modelling

2.1 Assumptions

The following simplifying assumptions are adopted throughout the analytical model:

1. The core material is infinitely permeable ($\mu_r \to \infty$), so the core reluctance is negligible.

2. Inductance varies sinusoidally with rotor position.

3. Fringing flux at the air-gap is neglected.

4. The magnetic system is linear (no saturation).

5. Magnetic flux is uniformly distributed across the core cross-section.

2.2 Part (a) - Reluctance and Inductance Analysis

By using the assumptions given, reluctance of the system can be calculated by the below formula:

$$\mathcal{R} = \frac{l}{\mu A_{core}} \tag{1}$$

where l is the path length, μ is the permeability and A_{core} is the core cross-sectional area. As the assumption states, core has infinitely permeable which makes the core reluctance zero. Thus, total reluctance will only consider the air gap reluctance. As the air-gap changes with respect to position, there will be a minimum and a maximum reluctance value. In below, all the parameters to find the reluctance and the maximum and minimum reluctance values is calculated.

The core cross-sectional area is:

$$A_c = w_c \cdot d = 15 \times 10^{-3} \times 25 \times 10^{-3} = 375 \times 10^{-6}\,\mathrm{m}^2 \tag{2}$$

When the rotor tooth is fully aligned with the stator pole which is the minimum inductance position, the total air-gap path length consists of two single air-gaps:

$$l_{g,\min} = 2g = 2 \times 0.5 = 1\,\mathrm{mm} \tag{3}$$

When the rotor is in the fully unaligned position, the flux path must also traverse the additional 2 mm each side:

$$l_{g,\max} = 2g + 2 \times 2\,\mathrm{mm} = 1 + 4 = 5\,\mathrm{mm} \tag{4}$$

Substituting into Equation (1):

$$\mathcal{R}_{\min} = \frac{1 \times 10^{-3}}{4\pi \times 10^{-7} \times 375 \times 10^{-6}} = 2.12 \times 10^{6}\,\mathrm{H}^{-1} \tag{5}$$

$$\mathcal{R}_{\max} = \frac{5 \times 10^{-3}}{4\pi \times 10^{-7} \times 375 \times 10^{-6}} = 1.06 \times 10^{7}\,\mathrm{H}^{-1} \tag{6}$$

The self-inductance of the coil is related to the reluctance by:

$$L = \frac{N^2}{\mathcal{R}} \tag{7}$$

Therefore, by using Equation (7):

$$L_{\max} = \frac{300^2}{\mathcal{R}_{\min}} = \frac{90{,}000}{2.122 \times 10^6} \approx 42.41\,\text{mH} \tag{8}$$

$$L_{\min} = \frac{300^2}{\mathcal{R}_{\max}} = \frac{90{,}000}{1.061 \times 10^7} \approx 8.48\,\text{mH} \tag{9}$$

2.2.1 Inductance as a Function of Rotor Position

The question further asks for the variation of the inductance with respect to the position under the assumption of sinusoidal variation. Since the machine has two poles due to variable reluctance, one mechanical revolution will produce two electrical cycles. Assuming a sinusoidal variation of inductance with the electrical angle θ:

$$L(\theta) = L_{\text{avg}} + L_{\Delta} \cos(2\theta) \tag{10}$$

where:

$$L_{\text{avg}} = \frac{L_{\max} + L_{\min}}{2} = \frac{42.41 + 8.48}{2} \approx 25.45\,\text{mH} \tag{11}$$

$$L_{\Delta} = \frac{L_{\max} - L_{\min}}{2} = \frac{42.41 - 8.48}{2} \approx 16.96\,\text{mH} \tag{12}$$

The resulting inductance profile is plotted in Figure 1.

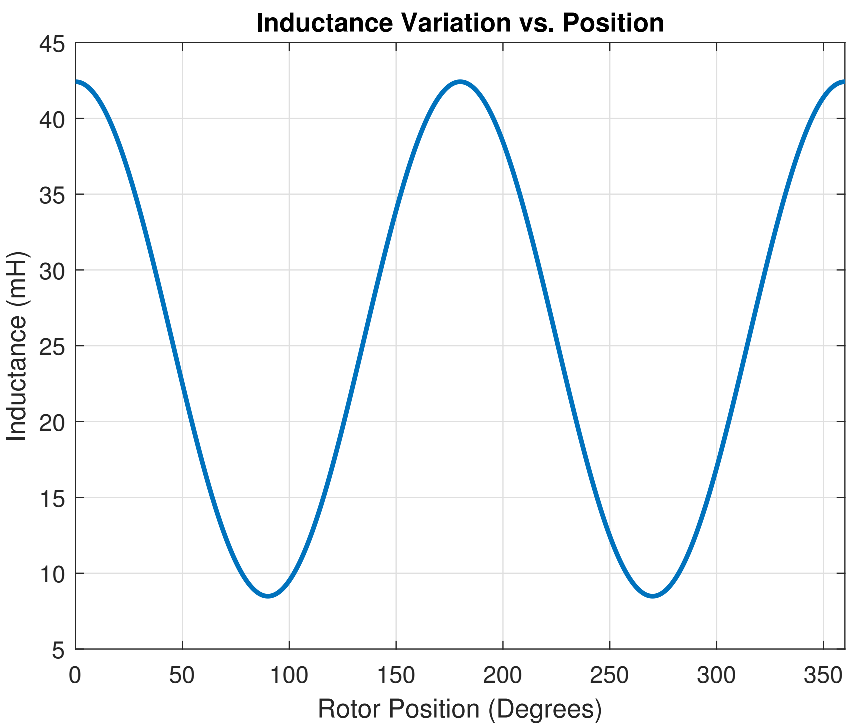

Figure 1: Inductance variation as a function of rotor position.

The inductance reaches its peak of 42.41 mH at the aligned positions (0° and 180°) and its minimum of 8.48 mH at the unaligned positions (90° and 270°).

2.2.2 Numerical Results Summary

Finally, numerical results for this part is summarized below.

Table 2: Analytical Reluctance and Inductance Results

Quantity	**Symbol**	**Value**
Minimum reluctance	$\mathcal{R}_{\text{min}}$	$2.122 \times 10^6\,\text{H}^{-1}$
Maximum reluctance	$\mathcal{R}_{\text{max}}$	$1.061 \times 10^7\,\text{H}^{-1}$
Maximum inductance	L_{max}	$42.41\,\text{mH}$
Minimum inductance	L_{min}	$8.48\,\text{mH}$

2.3 Part (b) -Torque under Constant DC Excitation

2.3.1 Torque Derivation

To derive the torque expression, one should make use of the stored magnetic energy on an inductor. The stored magnetic energy can be expressed in terms of flux linkage as:

$$W = \int_0^{\lambda} i(\lambda)\, d\lambda \tag{13}$$

For a linear magnetic system, the relationship between flux linkage and current is:

$$\lambda = L(\theta)\, i \tag{14}$$

Thus, the current can be written as:

$$i = \frac{\lambda}{L(\theta)} \tag{15}$$

Substituting into the energy expression:

$$W = \int_0^{\lambda} \frac{\lambda}{L(\theta)}\, d\lambda \tag{16}$$

Since $L(\theta)$ is independent of λ (linear system), it can be treated as constant:

$$W = \frac{1}{L(\theta)} \int_0^{\lambda} \lambda\, d\lambda \tag{17}$$

$$W = \frac{1}{L(\theta)} \cdot \frac{1}{2}\lambda^2 \tag{18}$$

Substituting $\lambda = L(\theta)i$:

$$W = \frac{1}{2}L(\theta)i^2 \tag{19}$$

The electromagnetic torque is obtained from:

$$T = \frac{dW}{d\theta} \tag{20}$$

$$T = \frac{d}{d\theta}\left(\frac{1}{2}L(\theta)i^2\right) \tag{21}$$

For constant current excitation:

$$T(\theta) = \frac{1}{2}i^2\frac{dL(\theta)}{d\theta} \tag{22}$$

Inductance variation is defined in the previous question. By taking the derivative, one can obtain the inductance variation as follows:

$$L(\theta) = L_{\text{avg}} + L_{\text{amp}}\cos(2\theta) \tag{23}$$

$$\frac{dL}{d\theta} = -2L_{\text{amp}}\sin(2\theta) \tag{24}$$

By inserting the inductance variation into Equation (22), the final torque equation becomes:

$$T(\theta) = -i^2 L_{\text{amp}}\sin(2\theta) \tag{25}$$

2.3.2 Torque as a Function of Rotor Position

The torque profile is plotted in Figure 2 by using the derived equation.

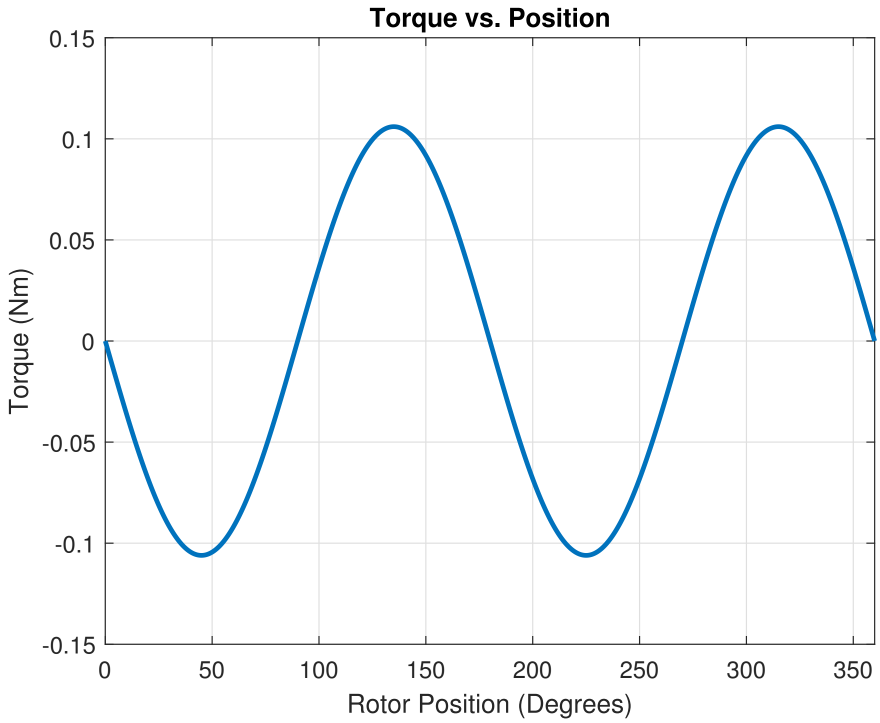

Figure 2: Electromagnetic torque as a function of rotor position under constant DC excitation ($I = 2.5\,\mathrm{A}$).

2.3.3 Results Summary

Table 3: Torque Results (Constant DC Excitation)

Quantity	**Symbol**	**Value**
Maximum torque	$T_{\max}$	$0.11\,\mathrm{Nm}$
Minimum torque	$T_{\min}$	$-0.11\,\mathrm{Nm}$

The torque is sinusoidal at twice the rotor frequency. It is positive in the regions where inductance is increasing (rotor moving towards alignment) and negative where inductance is decreasing. Under constant DC excitation, the average torque over a full revolution is zero, since the positive and negative half-cycles cancel.

2.4 Part (c) - Non-Linear Effects and Model Improvements

2.4.1 Fringing Flux

In the simple analytical model, flux is assumed to flow only through the face of the stator pole directly across the air-gap, ignoring the flux that flows outward around the pole edges. In practice, this fringing flux can significantly increase the effective cross-sectional area, particularly when the air-gap is large. One common approach to consider the fringing effects is to add the gap length to each linear dimension of the pole face to obtain an effective area:

$$A_{\text{eff}} = (w_c + g)(d + g) \tag{26}$$

This correction, as detailed in [1], is applied separately for each rotor position and reduces the reluctance, raising the predicted inductance closer to FEA results.

2.4.2 Non-Uniform Flux Distribution

The assumption of uniform flux distribution across the core is valid when the core is far from saturation conditions or the dimensions are such that the flux path is relatively uniform. A more accurate treatment to include would divide the core and air-gap regions into parallel flux tubes and solve for the flux in each tube independently, accounting for the variation of permeability in a saturating core. This would require numerical integration across the core cross-section and is more complex, but it would yield a more accurate prediction of inductance and torque, especially near saturation.

2.4.3 Saturation and Non-Linear Permeability

When the flux density in the core approaches the saturation flux density ($B_{\text{sat}} \approx 1.5\,\text{T}$ for typical electrical steel), the incremental permeability drops sharply. In this regime, the inductance is no longer constant for a given position. It becomes a function of both θ and I. For this condition, the co-energy must be computed numerically from the magnetisation curve:

$$W'(I, \theta) = \int_0^I \lambda(i, \theta)\, di \tag{27}$$

and the torque obtained as $T = \partial W' / \partial \theta \big|_{I=\text{const}}$.

To calculate this effect, the B-H curve of the selected lamination material to be included in the analysis.

3 Q2 - 2D FEA Modelling (Linear Materials)

In this section, the variable reluctance machine will be modelled in a 2D finite element analysis (FEA) software using linear material properties. The results will be compared with the analytical model derived in Q1 to validate the assumptions and identify any discrepancies. Main outputs from the FEA simulations will include flux density vectors, inductance values at different rotor positions, stored magnetic energy, and torque profiles. MAchine is generated in ANSYS Maxwell software by using the geometry provided on the course GitHub page. As the core material, M-19 electrical steel is selected due to its typical use in such applications and low eddy current losses. Material properties of the M-19 steel is available on ANSYS and the properties are presented in the appendix. In the linear part of the B-H curve, material has the relative permeability of $\mu_r \approx 5000$ which is used in the linear FEA simulations. Results for the analysis will be presented in the following subsections. Simulation model is presented in Figure 3.

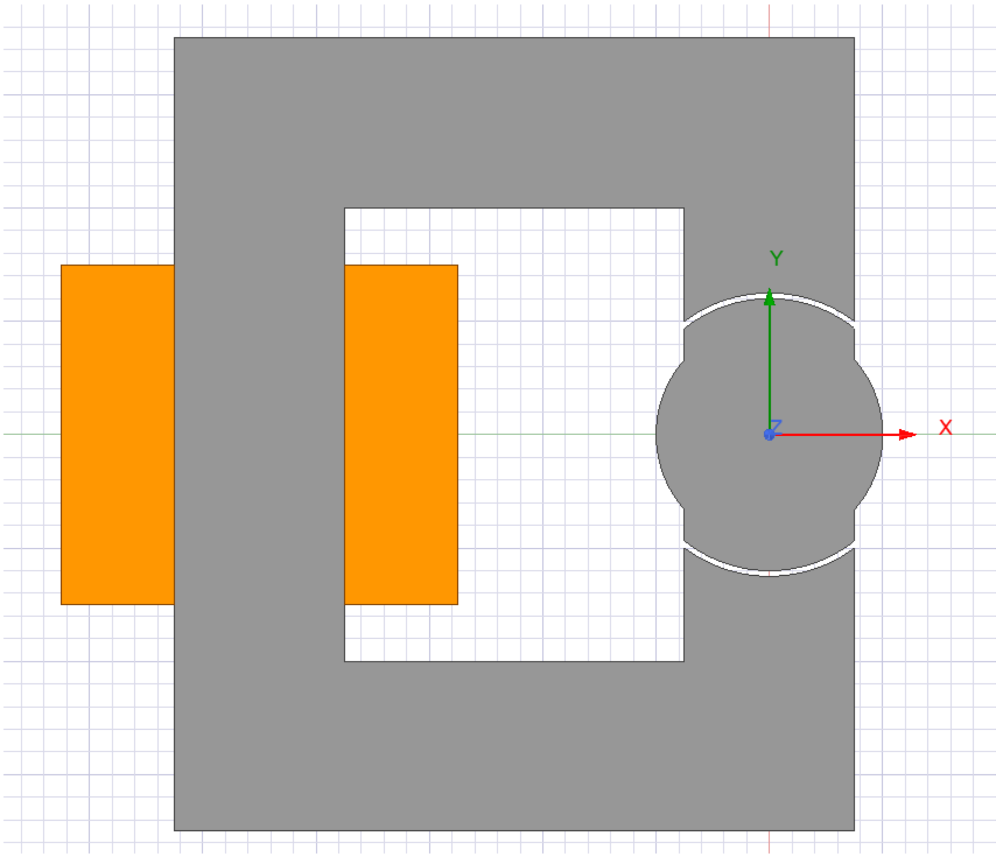

Figure 3: 2D FEA model of the variable reluctance machine in ANSYS Maxwell.

3.1 Part (a) - Flux Density Vectors at Different Rotor Positions

In this part, flux density vectors are plotted for three different rotor positions: aligned (0°), intermediate (45°), and unaligned (90°). The flux density distribution will illustrate how the magnetic field lines concentrate in the aligned position and spread out in the unaligned position, confirming the assumptions made in the analytical model. The flux density plots will be presented in Figures 4, 5, and 6.

As can be seen from the flux density plots, the flux lines are highly concentrated at the starting position (0°) where the rotor tooth is fully aligned with the stator pole, resulting in a high flux density. At the intermediate position (45°), the flux lines begin to spread out, indicating a decrease in flux density.

Finally, at the unaligned position (90°), the flux lines are widely spread, confirming the significant reduction in flux density and validating the assumptions of the analytical model. As the air gap increases with the rotor position, the reluctance increases which decreases the flux density while increasing the fringing fluxes across the air gap. These fringing fields can be observed in the flux density plots.

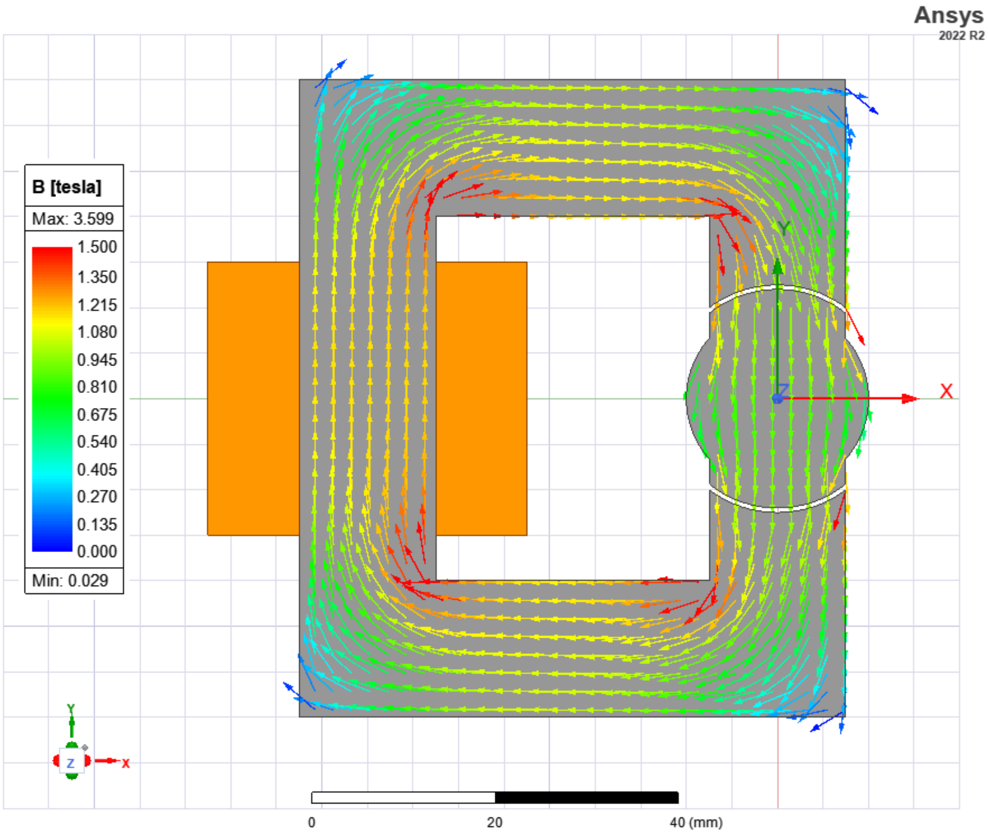

Figure 4: Flux density at 0°

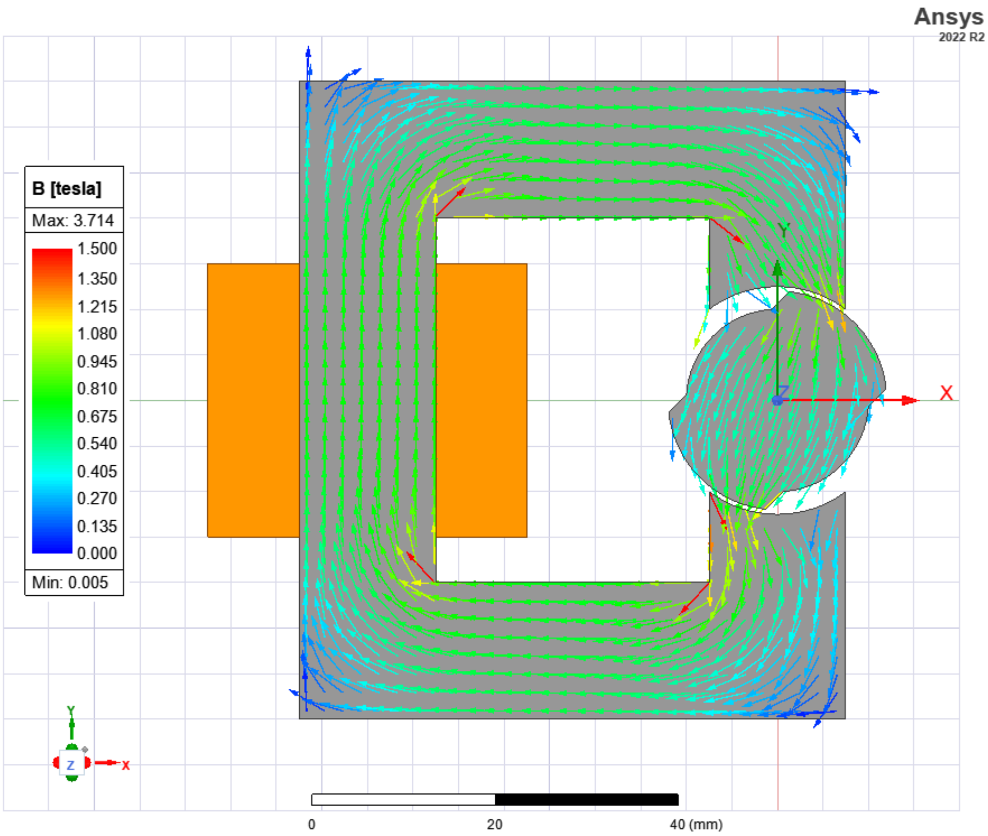

Figure 5: Flux density at 45°

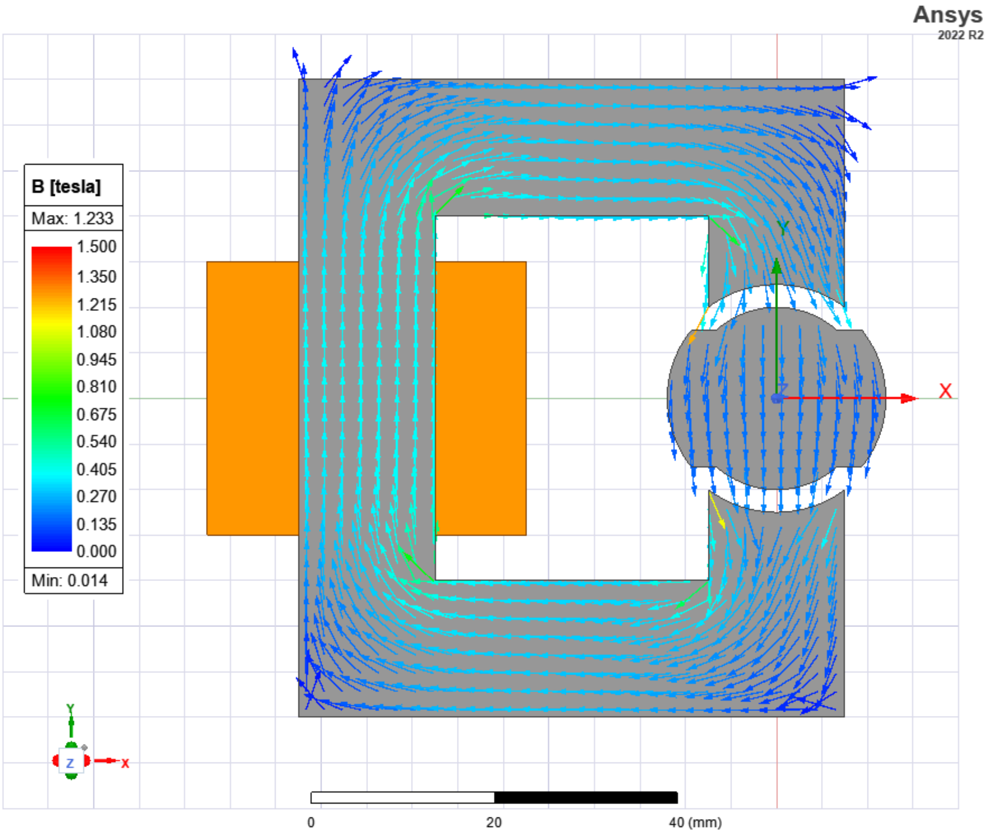

Figure 6: Flux density at 90°

Table 4: Inductance Values from FEA

Rotor Position	Inductance (mH)
0°	51.8
45°	33.1
90°	16.7

3.2 Part (b) - Inductance and Stored Energy

For the same three rotor positions, the inductance values and stored magnetic energy will be calculated from the FEA results. The inductance can be obtained from ANSYS itself by assigning the appropriate inductance parameters as the matrix, while the stored energy can be calculated using the co-energy method using Equation (19). These values will be compared with the analytical predictions to assess the accuracy of the analytical model and identify any discrepancies due to the assumptions made. Inductance results will be tabulated in Table 4 and the stored energy values will be presented in Table 5.

Table 5: Stored Energy Values from FEA

Rotor Position	Stored Energy (J)
0°	163.1
45°	104.1
90°	52.2

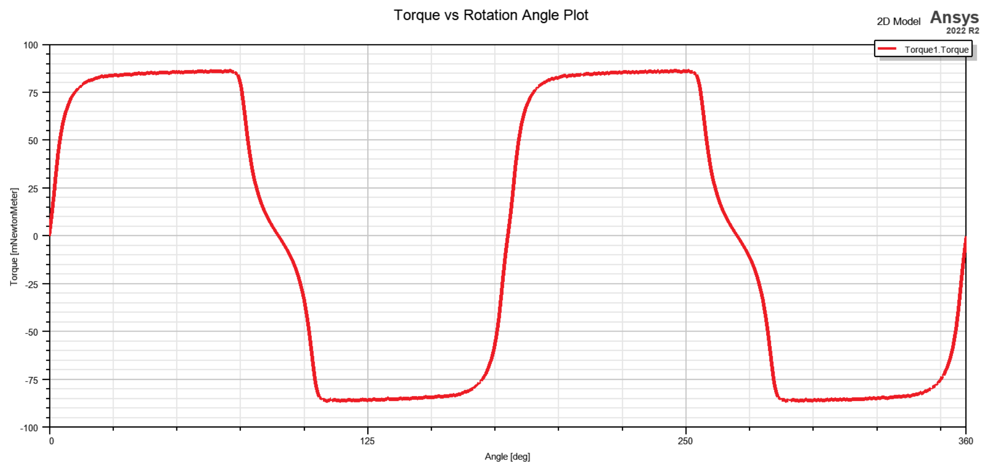

Figure 7: Torque variation with respect to rotor angle from FEA simulations.

3.3 Part (c) - Torque Variation with respect to Rotor Angle and Comparison with Analytical Model

In FEA, rotor is rotated in small increments of the angle (e.g., every 0.5°) and the torque is obtained at each position. The resulting torque profile is plotted as a function of rotor angle and presented in Figure 7.

4 Q3 - 2D FEA Modelling (Non-Linear Materials)

This section will be completed after the non-linear FEA simulations are performed.

4.1 Parts (a, b, c) - Flux Density, Inductance, Stored Energy, Torque

4.2 Part (d) - Comparison: Linear vs. Non-Linear, Fringing and Saturation Effects

5 Q4 - Control Method for Continuous Rotation

This section will be completed.

6 Conclusion

The analytical model of the variable reluctance machine was developed under linearised assumptions. The maximum and minimum inductances were found to be 42.41 mH and 8.48 mH, corresponding to the aligned and unaligned rotor positions respectively. The peak torque under 2.5 A DC excitation is ±0.106 Nm, varying sinusoidally at twice the rotor frequency.

FEA simulations (Q2, Q3) and the control strategy (Q4) will be added as the project progresses.

A Material Properties

This appendix will list the B-H curve data and material properties of the electrical steel lamination selected for the FEA simulations in Q2 and Q3.

B MATLAB Source Code

The MATLAB script used for the analytical calculations and plots in Q1 is provided below.

References

[1] W. G. Hurley and W. H. Wölfle, *Transformers and Inductors for Power Electronics: Theory, Design and Applications*, 1st ed. Chichester, UK: John Wiley & Sons, 2013. ISBN: 9781118544679.